Anway S. Pimpalkar

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EDUCATION

Harvard University, Cambridge, MA — Starting 2025
PhD, Bioengineering
Advisor: Shriya S. Srinivasan, PhD

Johns Hopkins University, Baltimore, MD — **GPA:** 4.00 / 4.00 — 2023 – 2025
MSE, Biomedical Engineering (Neuroengineering Focus)
Thesis: Bridging Perception and Performance: Generating, Calibrating, and Evaluating Multimodal Sensory Stimuli
Advisor: Jeremy D. Brown, PhD

College of Engineering Pune, Pune, India — **GPA:** 8.24 / 10.00 — 2019 – 2023
BTech, Electronics and Telecommunication Engineering

AWARDS AND HONORS

Harvard SEAS Prize Fellowship ($281,796 • 6-yr tuition and 2-yr stipend) • Harvard University — 2025 –
National Defense Science and Engineering Graduate Fellowship ($277,484 • 3-yr tuition and stipend) • DOD — 2025 –
Best Student Paper Finalist • International Conference on Rehabilitation Robotics (ICORR / RehabWeek) — 2025
National Institute on Aging (NIA) Start-Up Challenge Finalist ($10,000) • NIH — 2025
Best Paper Award • International Conference on Biomedical Engineering Science and Technology, India — 2023
Mitacs Globalink Award ($9,915) • Dalhousie University, Canada — 2022
Best Project Award • IEEE National Project Competition, India — 2021

RESEARCH EXPERIENCE

Johns Hopkins University • Haptics and Medical Robotics Lab ↗ — Baltimore, MD
Graduate Research Assistant • Advisor: Jeremy D. Brown, PhD ↗ — Jan 2024 –

Multimodal Visual-Haptic Neurorehabilitation for Post-Stroke Finger Dexterity

- Building a virtual task-based rehabilitation device to assess and enhance post-stroke finger dexterity, recording 3D isometric forces at 10kHz 2mN resolution using Python, Teensy MCU. Integrating voice coil-based haptic feedback with visual feedback led to 27% improvement in motor outcomes in 10 healthy participants compared to visual feedback alone. Highlights the need for multimodal feedback in sensorimotor rehabilitation. Podium presentation at BMES 2024.
- Validated subjective personalization of haptic and visual feedback through cross-modal matching with 12 additional healthy participants, tailoring feedback to account for individual differences in sensory perception after stroke. First-author paper accepted at IEEE ICORR (RehabWeek) 2025, finalist for Best Student Paper award.

Pneumatactors: Soft Multimodal Haptic Interfaces

- Engineering "pneumatactors" – a novel pneumatic tactor haptic interface designed for co-located stimulation of fast and slow adapting mechanoreceptors, enhancing cognitive salience. 3D-printed with flexible TPU, TPE to match mechanical impedance of human skin for intuitive interaction. Designed transistor-based solenoid valve switches to deliver upto 20kPa pressure and 0-143 Hz frequency, outperforming existing methods in the lower bandwidth. Integrating into prosthetic hand to assess perceptual benefits. Demonstrated live at IEEE Haptics Symposium.

Harvard University • Biohybrid Organs and Neuroprosthetics Lab ↗ — Cambridge, MA
Research Fellow • Advisor: Shriya S. Srinivasan, PhD ↗ — May 2024 – Aug 2024

Peripheral Nerve Stimulation to Modulate Cancer Tumor Metastasis

- Designed over-the-skin peripheral nerve stimulation paradigms to modulate melanoma tumor metastasis in mice models, leveraging modified TENS stimulators. Conducted pilot studies on 15 mice at varying stimulation frequencies, uncovering a frequency-dependent relationship between nerve stimulation and tumor innervation, confirmed through histology. Demonstrated a novel, non-invasive, and accessible alternative to invasive treatments for tumor control.
- Investigated mechanisms underlying glioblastoma multiforme interactions with the nervous system, targeting parasympathetic pathways to disrupt tumor-driven plasticity supporting growth. Designed, prototyped, and assembled

RF-powered wireless vagus nerve PCB stimulators (⌀13 mm) capable of delivering up to 1 kHz monophasic pulses using LDOs and oscillators in freely moving rats, enabling untethered power harvesting and stimulation.

Surgical Approach for Treating Stress Urinary Incontinence

- Innovated and tested a fully endogenous muscle graft-based surgical technique for treating stress urinary incontinence, featuring a 30-minute procedure with potential for fully laparoscopic implementation and superior efficacy compared to urethral devices. Conducted pilot surgeries on 5 rats, synchronizing bladder pressure, flow, and tension using a custom-designed PCB to assess and analyze urodynamic parameters. Results demonstrated therapeutic potential.

IIT Bombay • Human Motor Neurophysiology and Neuromodulation Lab ↗ — Mumbai, India

Research Intern • Advisor: Nivethida T, PhD ↗ — Jan 2023 – Aug 2023

Quantifying Prospective Component of Sense of Agency using Machine Learning

- Investigated pre-movement brain activity associated with intentional binding and the prospective Sense of Agency using a modified virtual Libet clock task, where voluntary movements were paired with probabilistic auditory feedback (0%, 50%, 75% probability). Applied analytical techniques in Python and MATLAB, including highly comparative time series analysis, spectral density estimation, and machine learning (SVCs, RFCs) to identify novel spatiotemporal EEG features.
- Discovered neural differences tied to action-outcome probabilities through high classification accuracies (74% to 100%) in distinguishing EEG signals across feedback blocks. Key features were localized to the pre-SMA and centroparietal regions, with distinct activity in the alpha and low gamma bands, in line with existing literature.
- Demonstrated the prospective Sense of Agency as a distinct neural phenomenon, providing new insights into the anticipatory neural mechanisms of agency and moving beyond traditional retrospective interpretations. Submitted research as Bachelor's Dissertation to the College of Engineering Pune.

College of Engineering Pune • Departments of Electronics and Applied Sciences ↗ — Pune, India

Undergraduate Researcher • Advisor: Mahesh H. Shindikar, PhD ↗ — Aug 2022 – Aug 2023

Deep Learning-Based MRI Skull Tissue Segmentation

- Developed and evaluated vanilla, residual, and dense 2D U-Net deep learning architectures for automated skull stripping in MRI neuroimaging, achieving 99.75% accuracy with the dense 2D U-Net on a test dataset. Demonstrated the advantages of dense interconnections, enabling feature reuse across layers in shallower models.
- Highlighted the utility of deep learning for critical neuroimaging tasks and research pipelines, particularly in workflows requiring proprietary software to independently skull strip MRIs from diverse scanners. Implemented data augmentation to enhance robustness to multi-scanner variability and improve model performance. Delivered an Podium presentation at ICBEST 2024, received the Best Paper Award among 50+ presentations.

Dalhousie University • Agricultural Mechanized Systems Group ↗ — Truro, Canada

Mitacs Globalink Research Intern • Advisor: Travis J. Esau, PhD ↗ — May 2022 – Aug 2022

Automating Harvesters to Reduce Operator Cognitive Load

- Automated wild blueberry harvesters, enabling automatic berry bin switching and dynamic harvester head height adjustment to reduce operator cognitive and physical load. Developed a camera-based sensing electronic control system with Time-of-Flight and RGB cameras to monitor bin fill levels, achieving ±10% accuracy in volumetric analysis and automatic bin switching. Integrated crop height data and GPS from the tractor's CAN bus for real-time harvester head height adjustment via linear actuators, ensuring optimal alignment with low-lying blueberry shrubs.
- Piloted the system over two successful harvest days, receiving positive operator feedback and harvesting precision.

Queliz Lifetech • Early Stage Medtech Startup ↗ — Pune, India

Research Intern • *Electronics Subsystem* — Jun 2021 – Sep 2021

Point-of-Care Device for Finger Dexterity Rehabilitation in Acute Burn Patients

- Engineered electronics control systems for an affordable, portable rehabilitation device for burn patients, focusing on customized, accessible therapy for limited MCP, DIP, and PIP finger joint mobility.
- Recorded precise finger flexion-extension paths using OpenCV-based computer vision (±2% path deviation) and integrated a closed-loop lead-screw actuation mechanism with adaptive feedback for accurate movement calibration.

CONFERENCE PRESENTATIONS † presenter

FULL-LENGTH PAPERS

C2 **A. S. Pimpalkar** †, A. M. West, J. Xu, and J. D. Brown, "Optimizing cross-modal matching for multimodal motor rehabilitation," *International Conference on Rehabilitation Robotics (ICORR / RehabWeek)*, May 2025. Best Student Paper Award Finalist. *Podium presentation.*

C1 **A. S. Pimpalkar** †, R. K. Patole, K. D. Kamble, and M. H. Shindikar, "Performance evaluation of vanilla, residual, and dense 2D U-Net architectures for skull stripping of augmented 3D T1-weighted MRI head scans," *International Conference on Biomedical Engineering Sci. & Tech.*, Feb 2023. Best Paper Award. *Podium presentation.* Paper ↗

ABSTRACTS AND SHORT ARTICLES

A7 **A. S. Pimpalkar** †, D. Rai, J. U. Bartels, J. Xu, and J. D. Brown, "Visual-haptic feedback enhances finger individuation in a virtual precision grip neurotraining task," *Biomedical Engineering Society (BMES),* Oct 2024. *Podium presentation.* Abstract ↗

A6 H. Song † et al. [including **A. S. Pimpalkar**], "Towards visual function restoration through photoacoustic stimulation," *Association for Research in Vision & Ophthalmology (ARVO)*, May 2024. *Podium presentation.* Abstract ↗

A5 **A. S. Pimpalkar** †, P. Ameta †, A. Dalia, and J. D. Brown, "Pneumatactor arrays for high frequency vibrotactile feedback," *IEEE Haptics Symposium*, Apr 2024. *Live demonstration and Work-in-Progress paper (2 presentations)*.

A4 **A. S. Pimpalkar** †, P. Ameta †, A. Dalia †, and J. D. Brown, "Pneumatactor arrays for high frequency vibrotactile feedback," *United States Senate AI Caucus: Robotics Demo Day*, Apr 2024. *Live demonstration.*

A3 C. Harris †, **A. S. Pimpalkar**, A. Aggarwal, P. Yang, X. Chen, C. Overby-Taylor, J. Greenstein, and R. D. Stevens, "Surgical risk prediction using an explainable deep learning approach applied to pre-operative 12-lead electrocardiograms," *Johns Hopkins Medicine & Engineering Research Retreat*, Feb 2024. *Poster presentation.* Poster ↗

A2 **A. S. Pimpalkar** †, R. Patole, and N. Thirugnanasambandam, "Demonstrating the prospective component of sense of agency using machine learning," *COEP ENTC BTech Project Symposium* , May 2023. *Poster presentation.* Poster ↗

A1 **A. S. Pimpalkar** †, R. Patole, K. Kamble, and M. Shindikar, "Evaluating U-Nets for skull stripping of augmented T1-weighted MRI scans," *No Garland Neuroscience Conference*, Feb 2023. *Podium and poster presentation.* Poster ↗

JOURNAL PUBLICATIONS

P3 **A. S. Pimpalkar**, A. Slepyan, and N. V. Thakor, "Vibrations at first contact encode object stiffness before grasp completion," *IEEE Sensors Letters* (in review), Nov 2024. Preprint: arXiv ↗

In Media: *Featured in New Scientist Magazine.* Article ↗

P2 C. Harris, **A. S. Pimpalkar**, A. Aggarwal, J. Yang, X. Chen, S. Schmidgall, S. Rapuri, J. Greenstein, C. Overby-Taylor, and R. D. Stevens, "Risk prediction for non-cardiac surgery using the 12-lead electrocardiogram: An explainable deep learning approach," *British Journal of Anesthesiology* (in review), Feb 2025. Preprint: medRxiv ↗

P1 **A. S. Pimpalkar** and D. Niture, "Towards contactless elevators with tinyML using person detection and keyword spotting," Jul 2024. Best Project Award. Preprint: arXiv ↗, Code ↗

TEACHING ENGAGEMENTS

Mechatronics, Johns Hopkins • Senior Level — Spring 2024

Teaching Assistant to Jeremy D. Brown, PhD

- Solved a 3-year challenge in real-time location polling of autonomous air hockey robots using CV-based ArUco tag detection and ZigBee mesh networking (open-source implementation ↗). Led labs on building autonomous robots. Worked in a mentorship role in Spring 2025.

Principles of the Design of Biomedical Instrumentation, Johns Hopkins • Graduate Level — Fall 2024

Teaching Assistant to Nitish V. Thakor, PhD

- Mentored labs on designing ECG, EMG, EEG recording circuits and projects including accessible gaming, wearables.

Haptic Interface Design for Human-Robot Interaction, Johns Hopkins • Graduate Level — Fall 2024

Class Researcher with Jeremy D. Brown, PhD

- Helped pilot an online learning platform to provide self-paced lectures incorporating live instructor interaction.

ENTREPRENEURIAL ACTIVITIES

Re-Kinesis • Early Stage Medtech Startup ↗ Baltimore, MD

Chief Executive Officer • NIH National Institute on Aging Start-Up Challenge Finalist Dec 2024 –

- Re-Kinesis is an AI-powered digital health platform transforming ordinary footwear into real-time biomechanical assessment tools. Using high-resolution plantar pressure sensing and machine learning, we provide clinical-grade mobility tracking, enabling real-time rehabilitation, injury prevention, and performance optimization.

INVITED TALKS

Haptics: Neural Basis, Applications, Future Avenues. **Johns Hopkins University, Sep 2024**. **Neural & Rehabilitation Engineering Course**, invited by Nitish V. Thakor, PhD.

SKILLS AND CERTIFICATIONS

Medical Research	Human participant studies (non-invasive devices), rodent studies (invasive devices, mice/rats), human perception experiment design, IRB protocols
Electronics	Analog and digital circuit design, microcontroller programming (Arduino/Teensy/ESP/STM), high-speed and high-current PCB design (KiCAD/Altium), closed-loop control systems, DAQ design, FPGA, RF design, precision soldering (THT/SMT), mechatronic integration, communication protocols
Programming & CS	Python, MATLAB, C/C++, data analysis, VR rendering, machine and deep learning, TinyML, TensorFlow (+lite/micro), computer vision, LaTeX
Fabrication & Design	CAD (Autodesk, Shapr3D), FDM 3D printing (rigid/flexible), SLA resin printing, silicone molding
Certifications ↗ (15 total courses)	**Harvard Online** • TinyML Certificate (3 courses), Python for Research, CS50 Web Development **MIT Online** • Aeronautics and Human Spaceflight **Johns Hopkins Online** • Human Subject Research, Neuroimaging, Neurohacking **DeepLearning.AI** • Deep Learning Specialization (5 courses) **AICTE** • Computer Science & Biology (faculty development program completed with permission)

ACADEMIC SERVICE

IEEE Haptics Symposium • Student Volunteer 2024

Canadian Society of Bioengineering Annual Meeting • Student Volunteer 2022

LEADERSHIP AND TEAMWORK

Johns Hopkins India Institute • Student Leader, Operations and Logistics Present

- Managing operations, logistics, sponsorships for university's first Hopkins India Conference, May 2025, Washington DC.

Johns Hopkins Biomedical Engineering • Master's Program Representative 2024

- Represented the department at over 7 information sessions and conferences, including BMES 2024, upon invitation.

COEP Impressions Cultural Festival • Head of Events (2021-22), Volunteer 2019 – 2022

- Led the planning and execution of 28 annual events with 10k attendees, managing a ₹2m budget and coordinated with an overall team of 126 members to execute concerts, competitions, and international celebrity engagements.
- Built and led a team of 17, working with two co-heads to build a platform from the ground up, fostering opportunities for budding artists across India and uniting them under a shared vision. Placed emphasis on unconventional art forms.

Rowing Team • COEP, Pune City • Competitive 1X, 2X, 4X Sculler 2020 – 2022

- Won multiple race events and an institutional award commending motivation and skill. Represented city at regional races.

COEP Mental Health & Wellness Center • Student Volunteer 2020 – 2022

COEP Data Science & AI Club • Research Lead (2020-21), Member 2020 – 2022